Module 3, Lesson 3.4: Daily Notes & Long-Term Memory

The Two Memory Systems

Your AI agent has two complementary memory systems that work together:

Daily Notes capture the immediate—what happened today, what you worked on, decisions made, and information encountered. They're like a journal, recording the raw material of your activities.

Long-Term Memory captures the enduring—what matters over time, patterns and preferences, key decisions and their reasoning. It's like your curated knowledge, refined and organized.

Understanding how these systems work together—and how to use them effectively—is the key to building an AI partner that gets smarter over time.

Daily Notes: Capturing the Now

What Are Daily Notes?

Daily notes are automatically created files in your `memory/` folder, one for each day you interact with your agent:

```
memory/
├── 2024-01-15.md
├── 2024-01-16.md
├── 2024-01-17.md
└── ...
```

Each file contains a record of that day's interactions, decisions, and important information.

What Goes in Daily Notes

Conversations

- Key topics discussed
- Decisions made
- Questions asked and answered
- Information exchanged

Work Done

- Tasks completed
- Progress on projects
- Files created or modified
- Problems solved

Information Captured

- Facts learned
- Resources discovered
- Contacts mentioned
- Ideas generated

Status Updates

- Project status changes
- Goal progress
- Blockers encountered
- Next steps identified

How Daily Notes Are Created

Your agent automatically creates and updates daily notes. You can influence what gets captured:

Explicit Capture: “Note this in my daily notes: I decided to focus on the Acme project this week and pause the website redesign.”

Implicit Capture: Your agent records significant interactions automatically. Important decisions, new information, and progress updates get logged.

Query Your Notes: “What did I work on yesterday?” “What decisions did I make last week?” “Show me my notes from January 10th.”

Daily Note Structure

While your agent handles the structure, understanding the format helps you work with it:

```
# Daily Note: 2024-01-15

## Summary
Brief overview of the day's key activities and outcomes.

## Conversations
Key discussion points and outcomes.

## Decisions Made
- Decision 1: [What] - Why: [Reasoning]
- Decision 2: [What] - Why: [Reasoning]

## Progress
### [Project 1]
- What was done
- Current status
- Next steps

### [Project 2]
- What was done
- Current status
- Next steps

## Information Captured
- [Facts, resources, contacts learned about]

## Tasks
- [x] Completed task
- [ ] Pending task
- [ ] New task identified

## Notes
[Running notes from the day]

## Tomorrow
- [Priority items for next session]
```

Working with Daily Notes

Session Start When you begin a session, your agent loads recent daily notes:

“Good morning! I’ve reviewed your recent activity. Yesterday you focused on the Acme project and made progress on the proposal. You have a call with Sarah today at 2 PM.

What would you like to work on?”

Session End When you’re done working, ask for a summary:

“Summarize what we accomplished today and add it to my daily notes.”

Mid-Session Capture During work, capture important moments:

“This is important—note that we decided to go with Option B for the Acme project because of cost and timeline.”

Retrospective Review Look back over your notes:

“Review my daily notes from last week and identify any patterns or recurring themes.”

Long-Term Memory: Capturing the Enduring

What Is Long-Term Memory?

Long-term memory lives in `MEMORY.md`—a curated file that persists across all sessions. Unlike daily notes which accumulate daily, long-term memory is refined and maintained.

What Belongs in Long-Term Memory

Permanent Facts

- Your goals and priorities
- Key relationships and contacts
- Important dates and deadlines
- Your preferences and patterns

Enduring Decisions

- Strategic choices and their reasoning
- Commitments you’ve made
- Boundaries you’ve established
- Principles you follow

Patterns and Insights

- What works for you
- What doesn’t work for you
- Lessons you’ve learned
- Observations about your work

Reference Information

- How you organize things
- Your systems and processes
- Key information you need regularly
- Context that should persist

What Doesn’t Belong in Long-Term Memory

Temporary Information

- Today’s weather
- What you had for lunch
- Temporary states that will change

Detailed Logs

- Every email sent
- Every file created
- Step-by-step records

Easily Retrieved Information

- Information already in your PARA structure
- Things documented elsewhere
- Redundant data

Managing Long-Term Memory

Adding to Memory

Explicit addition: “Remember this: I prefer to have my most important meetings in the morning, between 9 and 11 AM.”

Through conversation: “I’ve decided to focus on consulting for small businesses. This is a strategic shift—note it in my long-term memory.”

Querying Memory

“What do you know about my business focus?” “What are my current priorities?” “When is my biggest client’s deadline?”

Updating Memory

“Update my memory: I’ve shifted my target market from small businesses to mid-size companies.”

“Forget about my old project focus—I’ve changed direction.”

Reviewing Memory

“Read my MEMORY.md and summarize what you know about me.” “What should I add to my long-term memory based on recent activity?”

MEMORY.md Structure

Here’s a recommended structure for your long-term memory file:

```md
# MEMORY.md - Long-Term Memory

## Personal Context
### Identity
- Name: [Your name]
- Role: [Your professional identity]
- Location: [Where you are]
- Timezone: [Your timezone]

### Current Priorities
1. [Priority 1]
2. [Priority 2]
3. [Priority 3]

### Goals
**This Quarter**
- [Quarterly goals]

**This Year**
- [Annual goals]

**Long-Term**
- [Long-term vision]

## Work Context
### Business/Projects
- [Current business focus or key projects]
- [Important context about your work]

### Key Contacts
- **[Name]**: [Relationship, context, last interaction]
- **[Name]**: [Relationship, context, last interaction]

### Important Dates
- [Date]: [Event]
- [Date]: [Event]

## Preferences
### Communication
- [How you prefer to communicate]
- [Your communication style]

### Work Style
- [How you work best]
- [Your productive hours]
```

```
- [Your approach to tasks]

### Tools & Systems
- [Key tools you use]
- [How you organize your work]

## Patterns & Insights
### What Works
- [Approaches that have been successful]

### What Doesn't Work
- [Approaches to avoid]

### Lessons Learned
- [Important lessons from experience]

## Decisions Made
| Date | Decision | Reasoning |
|------|----------|-----------|
| [Date] | [Decision] | [Why you made this choice] |

## Current Status
### Projects
- [Project 1]: [Current status, next milestone]
- [Project 2]: [Current status, next milestone]

### Areas of Life
- Health: [Current focus]
- Finances: [Current focus]
- Career: [Current focus]
- Relationships: [Current focus]
- Home: [Current focus]

## To Add
- [Things to consider adding to memory]
```

The Memory Flow: From Daily to Long-Term

The real power comes from how these two systems interact.

Automatic Flow

Your agent can identify information in daily notes that belongs in long-term memory:

"I noticed you mentioned your new client three times this week. Should I add them to your long-term memory under Key Contacts?"

Manual Flow

You can explicitly move information:

"Take the decision about pricing from yesterday's notes and add it to my long-term memory under Decisions Made."

"Review this week's daily notes and extract anything that should go in long-term memory."

Review Flow

Periodic reviews surface valuable information:

Weekly Review: "This week you made progress on three projects, learned about [topic], and made a key decision about [subject]. What should I add to long-term memory?"

Monthly Review: "I've identified several patterns in your daily notes from this month: [patterns]. Should any of these be captured in long-term memory?"

Memory Maintenance Routines

Daily Practices

Morning

- Brief review of yesterday's notes
- Identify any items for long-term memory
- Set intentions for today's note-taking

During the Day

- Capture significant moments explicitly
- Note decisions and their reasoning
- Record information worth remembering

End of Day

- Review what was captured
- Highlight important items
- Flag items for long-term memory

Weekly Practices

Memory Review Session (15-20 minutes)

1. **Review Daily Notes**
    - Read through the week's notes
    - Identify patterns and themes
    - Note significant decisions
2. **Extract to Long-Term Memory**
    - Add new contacts
    - Update project status
    - Record important decisions
3. **Clean Up**
    - Archive old items
    - Update outdated information
    - Remove redundant entries
4. **Plan for Next Week**
    - What do you want to remember?
    - What do you want your agent to track?

Monthly Practices

Deep Memory Audit (30-60 minutes)

1. **Read All of MEMORY.md**
    - Is everything still accurate?
    - Is anything missing?
    - Is anything outdated?
2. **Review Patterns**
    - What patterns have emerged this month?
    - What insights should be captured?
    - What lessons were learned?
3. **Archive and Condense**
    - Archive completed projects
    - Condense verbose sections
    - Reorganize if needed
4. **Set Intentions**
    - What do you want your agent to focus on?
    - What information will be important next month?

Advanced Memory Techniques

Technique 1: Memory Tags

Use tags to categorize memories for easy retrieval:

```
[PROJECT:Acme] Client prefers email over phone
[LEARNING] Batch similar tasks for efficiency
[PREFERENCE] I'm most creative in the morning
[CONTACT] Sarah mentioned she's allergic to shellfish
```

Then query by tag: “Show me all PROJECT:Acme memories.” “What have I learned this month?”

Technique 2: Memory Summaries

Periodically create summaries:

```
## January 2024 Summary

### Key Accomplishments
- Launched consulting website
- Signed first client (Acme Corp)
- Completed AI automation course

### Key Decisions
- Pivoted from product to service offering
- Set pricing at $X/hour

### Patterns Noticed
- Most productive on Tuesday mornings
- Clients respond best to case studies

### Goals for February
- [Next month's goals]
```

Technique 3: Contextual Memory

Link memories to contexts:

```
### Context: Client Proposals
When writing proposals:
- Include case studies (proven effective)
- Keep under 5 pages (client feedback)
- Lead with outcomes, not process
- Sarah prefers PDF format
```

When working on proposals, your agent can reference this context.

Technique 4: Progressive Memory Building

Build memory over time through refinement:

Day 1: “I’m working with Acme Corp.” **Day 5:** “Acme Corp is a 25-person manufacturing company.” **Day 10:** “Acme Corp’s main pain point is inventory management.” **Day 15:** “Acme Corp’s decision maker is Sarah, the COO.”

Each interaction adds depth. Your agent builds a progressively richer understanding.

Memory Troubleshooting

“My agent doesn’t remember things between sessions”

Check:

- MEMORY.md exists and is readable
- Daily notes are being created
- Agent is loading recent notes at start

Solution:

- Explicitly tell your agent to remember: “Remember this…”
- Check your configuration for memory settings
- Verify files aren’t being overwritten

“My agent remembers too much irrelevant stuff”

Check:

- What’s in MEMORY.md?
- What’s in recent daily notes?

Solution:

- Review and clean MEMORY.md
- Be more selective about what you ask to remember
- Prune daily notes to key information

“My agent doesn’t know something I told it”

Check:

- Was it actually recorded?
- Was it in a recent session or older?
- Is it in MEMORY.md or just daily notes?

Solution:

- Explicitly add important things to long-term memory
- Reference relevant context: “Remember when we discussed…”
- Review and consolidate memory periodically

“My agent gives outdated information”

Check:

- When was MEMORY.md last updated?
- Are there conflicting entries?

Solution:

- Update outdated entries: “Forget my old pricing. My new pricing is…”
- Regular memory audits
- Explicitly note when information changes

The Payoff: Memory That Compounds

The beautiful thing about AI memory is that it compounds. Every day you work with your agent:

- It knows you better
- It remembers more context
- It requires less explanation
- It provides more relevant help

Month 1: You explain everything. Month 3: You explain most things. Month 6: You explain only what's new. Month 12: Your agent anticipates your needs.

This compounding effect is why memory isn't just nice to have—it's the foundation of true AI partnership.

Coming Up: Module 4

You've now built a complete memory system for your AI agent. In Module 4, we'll put it to work with automation workflows that actually pay—email management, calendar intelligence, content creation, and more.

Key Takeaways

1. **Daily notes capture the immediate** - Raw material of your activities
2. **Long-term memory captures the enduring** - Curated knowledge that persists
3. **The two systems flow together** - Daily informs long-term, long-term informs daily
4. **Regular maintenance keeps memory valuable** - Weekly reviews, monthly audits
5. **Memory compounds over time** - Every session builds on previous ones